

Build Advanced and Highly Reliable Industrial Ethernet Communication Systems



ISM7012D-P Managed Industrial Ethernet Switch

· L2+ Managed

· 4×SFP + 8×GT

· Ring Redundancy

· IP40

1. Product Overview

The ISM series of managed industrial Ethernet switches is purpose-built for harsh environments in power, transportation, and industrial control, delivering reliable operation under extreme temperatures.

Product Positioning: Build advanced and highly reliable industrial Ethernet communication systems

Product Highlights:

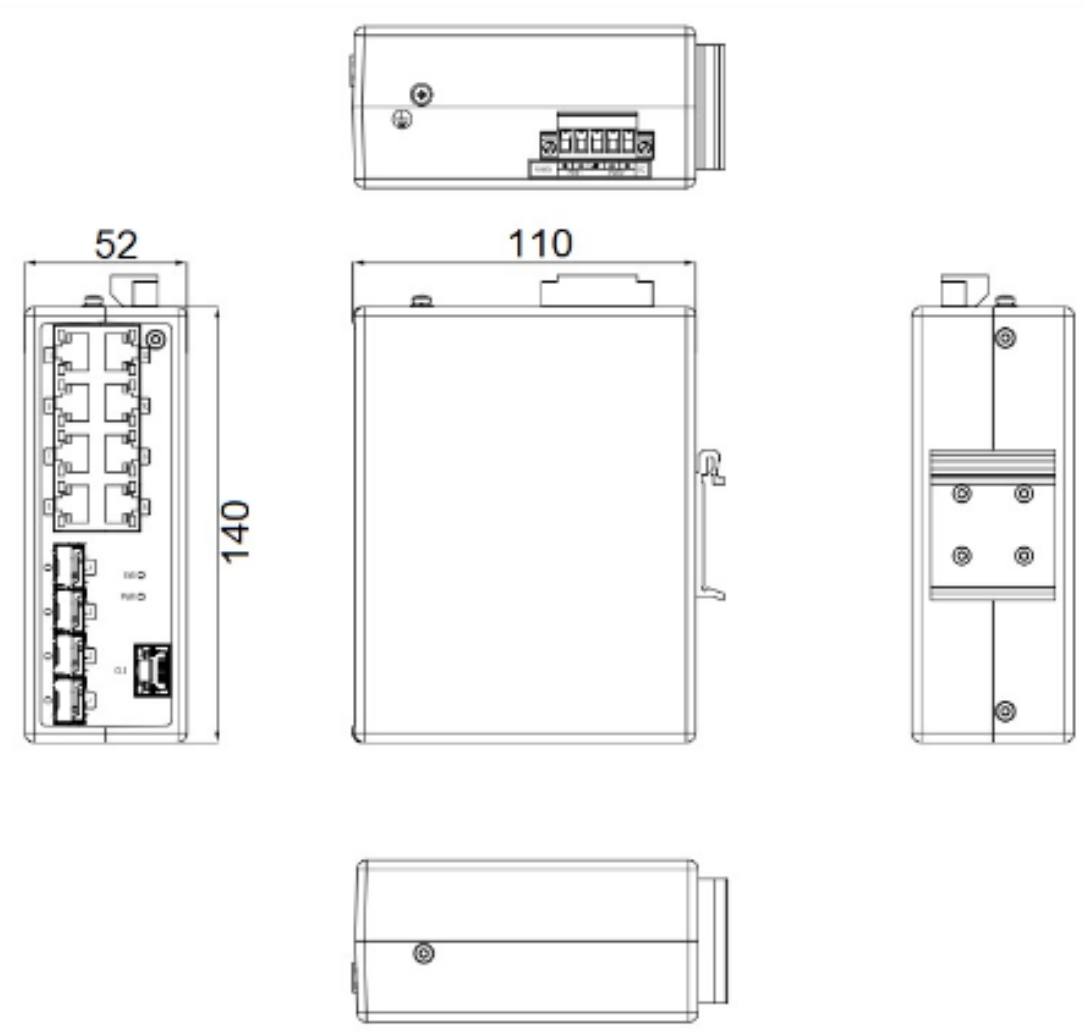
- **Robust Industrial Design:** Rugged metal casing with protective coating, fanless, IP40, wide-temperature operation
- **Redundant Power Supply:** Dual input power design with wide voltage input for stable communication
- **Ring Redundancy:** STP / RSTP / MSTP ring redundancy protocols for flexible network topologies
- **Easy Management:** SNMPv1/v2c/v3 and RMON management, plug-and-play, quick DIN-rail deployment
- **High Reliability:** MTBF over 35 years for long-term mission-critical operation

Key Technical Specifications

Item	Specification
Dimensions (W × D × H)	52 × 140 × 110 mm
Weight	0.7 kg
Mounting	DIN-rail mounting
Ports	4 × 1000Base-X SFP + 8 × 10/100/1000Base-T
Switching Capacity	68 Gbps backplane bandwidth
Power Supply	18–60 V DC, redundant dual input
Operating Temperature	-40 °C ~ +75 °C

Item	Specification
Ingress Protection	IP40

2. Product Dimensions



ISM7012D-P (52 × 140 × 110 mm)

Note:

- 1. All dimensions are in millimeters (mm).
- 2. All dimensions are approximate and **for reference only**.
- 3. The dimensions shown in the figure **shall not be used for production or processing**.
- 4. Dimensions must comply with part and manufacturing tolerance requirements.
- 5. Dimensions are subject to change without notice.

3. Hardware Specifications

Category / Parameter	Specification
Hardware Performance	
Backplane Bandwidth	68 Gbps
Forwarding Mode	Store-and-Forward
MAC Address Table	16K
Packet Buffer	4 Mbits
Forwarding Rate	148,800 pps @ 100M ports; 1,488,000 pps @ 1000M ports
Interfaces	
SFP Port	4 × 1000Base-X SFP (SFP module not included)
Ethernet Port	8 × 10/100/1000Base-T RJ45
Console Port	1 × Management Serial CLI Port
Power	
Input Voltage	18–60 V DC, redundant dual input
Overcurrent Protection	Supported
Reverse Polarity Protection	Supported
Power Consumption	10 W
Mechanical	

Category / Parameter	Specification
Dimensions (W × D × H)	52 × 140 × 110 mm
Weight	0.7 kg
Enclosure	Fully enclosed seamless metal enclosure
Ingress Protection	IP40
Cooling Method	Fanless
Mounting	DIN-rail mounting
Environmental	
Operating Temperature	-40 °C ~ +75 °C
Storage Temperature	-40 °C ~ +85 °C
Humidity	5 ~ 95 % RH (non-condensing)
EMC	
EMI	FCC 47 CFR Part 15 Class A; EN55022 Class A; EN55035 Class A
ESD Immunity	IEC (EN) 61000-4-2, Level 4
Radiated Electric Field	IEC (EN) 61000-4-3, Level 3
Electrical Fast Transient/Burst Immunity	IEC (EN) 61000-4-4, Level 4
Surge	IEC (EN) 61000-4-5, Level 4
Conducted Immunity	IEC (EN) 61000-4-6, Level 3

Category / Parameter	Specification
Power Frequency Magnetic Field	IEC (EN) 61000-4-8, Level 5
Shock and Vibration	
Shock Resistance	IEC 60068-2-27
Drop	IEC 60068-2-31
Vibration Resistance	IEC 60068-2-6
Quality Assurance	
Warranty Period	5 years
MTBF	35 years

4. Software Specifications

Category / Parameter	Specification
Redundancy	
Ring Protocols	STP, RSTP, MSTP
Link Aggregation	Port Trunking (IEEE 802.3ad)
Flow Control	IEEE 802.3x
VLAN	
VLAN	Port-based VLAN, IEEE 802.1Q VLAN, GVRP
QoS	
Priority & Queuing	IEEE 802.1P/1Q, TOS/DiffServ
Multicast	

Category / Parameter	Specification
Multicast Filtering	IGMP Snooping, GMRP/GARP
Routing	
IP Routing	Static routing, RIP v2, OSPFv2, BGP, VRRP
Network Security	
Authentication	IEEE 802.1x, RADIUS, SSH2
Network Management	
Management Mode	Browser (Web), Serial Port (CLI), Telnet
SNMP	SNMPv1, SNMPv2c, SNMPv3, MIB-II, SMIV2, RMON, Ping MIB (RFC 2925)
Time Synchronization	SNTP, NTP
Diagnostics	
Diagnostic Mode	Indicator light, Journal File (Syslog), RMON, Port Mirroring, TRAP

5. Ordering Information

Model Code

Model code: ISM7012D-P-<SFP>-<GT>-<Temp>

<SFP>: Number and type of SFP fiber ports

<GT>: Number and type of Gigabit RJ45 ports

<Temp>: Operating temperature grade

Product Models

Model	Region	Description
ISM7012D-P-4GSFP-8GT-24	Global	12-Port Layer 2+ Managed Industrial Switch. 4 × 1000Base-X SFP Ports (SFP module not included), 8 × 10/100/1000Base-T Ports, 1 × Management Serial CLI Port. IP40 Protection Class, Operating Temperature -40 °C ~ +75 °C, Isolated Dual 18–60 V DC Power Inputs.

6. Contact Us

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